PROJECT SYNOPSIS

Water Quality Classification and Contamination Risk Prediction

Using Machine Learning on Physicochemical Water Quality Indicators

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1. Introduction

Access to safe drinking water is a basic requirement for public health, yet waterborne contamination remains a major cause of disease worldwide. Conventional laboratory testing of water samples is accurate but slow, costly and often unavailable in remote areas. Machine learning offers a faster alternative: by learning patterns from measured physicochemical parameters, a model can classify a water sample as safe or unsafe and flag contamination risk early, supporting timely decisions by water authorities and communities.

2. Problem Statement

Determining whether water is potable requires evaluating several interacting parameters such as pH, hardness, dissolved solids, disinfectant residuals and organic content. Manually checking each parameter against guideline limits is time-consuming and does not capture combined effects. This project aims to build a data-driven classification system that predicts water potability and estimates contamination risk from routine water quality measurements.

3. Objectives

- To explore and understand the distribution and relationships of water quality parameters.
- To preprocess the data, including handling missing values, outliers and class imbalance.
- To build and compare machine learning classifiers (e.g. Logistic Regression, Decision Tree, Random Forest, SVM, Gradient Boosting/XGBoost) for potability prediction.
- To evaluate models using accuracy, precision, recall, F1-score, ROC-AUC and confusion matrices.
- To derive a contamination risk score (probability of non-potability) and identify the most influential parameters.

- To present the results through clear visualisations and, optionally, a simple prediction interface.

4. Dataset Description

The project uses the Water Potability dataset (water_potability.csv), containing 3,276 water samples described by nine physicochemical features and one binary target variable, Potability. A summary of the variables is given in Table 1.

Table 1: Dataset features and summary statistics

Feature	Description	Range (min–max)	Mean
ph	pH of water (acid–base balance)	0.00 – 14.00	7.08
Hardness	Capacity of water to precipitate soap (mg/L)	47.43 – 323.12	196.37
Solids	Total dissolved solids, TDS (ppm)	320.94 – 61,227.20	22,014.09
Chloramines	Disinfectant level (ppm)	0.35 – 13.13	7.12
Sulfate	Dissolved sulfate (mg/L)	129.00 – 481.03	333.78
Conductivity	Electrical conductivity (µS/cm)	181.48 – 753.34	426.21
Organic_carbon	Total organic carbon (ppm)	2.20 – 28.30	14.28
Trihalomethanes	Disinfection by-products (µg/L)	0.74 – 124.00	66.40
Turbidity	Cloudiness of water (NTU)	1.45 – 6.74	3.97
Potability	Target: 1 = potable, 0 = not potable	0 / 1	0.39

Key observations from the initial inspection of the data:

- Class distribution: 1,998 samples (61%) are non-potable and 1,278 samples (39%) are potable, indicating a moderate class imbalance.
- Missing values: ph has 491 missing entries (15.0%), Sulfate has 781 (23.8%) and Trihalomethanes has 162 (4.9%). All other columns are complete.
- Weak linear relationships: the correlation of each feature with Potability is very low (absolute value below 0.04), suggesting that non-linear and multivariate models will be needed and that high accuracy should not be assumed.
- Scale differences: Solids is in the tens of thousands while Chloramines and Turbidity are in single digits, so feature scaling is required for distance- and gradient-based models.
- Some values lie outside realistic ranges (e.g. pH of 0 or 14), which need to be checked as possible outliers.

5. Methodology

The work will follow a standard data science workflow:

- Data collection and understanding: load the dataset and review its structure, summary statistics and quality.

- Data preprocessing: impute missing values (median or KNN imputation), treat outliers, standardise features and split the data into training and testing sets using stratified sampling.
- Exploratory data analysis: histograms, box plots, correlation heatmap and class-wise comparison of parameters.
- Handling class imbalance: class weights or resampling (e.g. SMOTE) applied to the training data only.
- Model building: train several classifiers and tune hyperparameters using cross-validation and grid/random search.
- Evaluation: compare models on accuracy, precision, recall, F1-score and ROC-AUC, with emphasis on recall for the non-potable class because unsafe water misclassified as safe is the costlier error.
- Contamination risk prediction: convert predicted probabilities into Low, Medium and High risk categories and analyse feature importance (e.g. permutation importance or SHAP).

6. Tools and Technologies

Category	Tools
Programming language	Python 3
Libraries	Pandas, NumPy, Scikit-learn, XGBoost, imbalanced-learn, Matplotlib, Seaborn
Environment	Jupyter Notebook / Google Colab
Optional deployment	Streamlit or Flask for a prediction interface

7. Expected Outcomes

- A cleaned and well-documented dataset ready for modelling.
- A comparison of multiple classifiers identifying the best-performing model for potability prediction.
- A contamination risk score and risk categories for individual water samples.
- Insights into which water quality parameters contribute most to potability decisions.

8. Applications and Significance

The system can support municipal water departments, rural water supply programmes and environmental monitoring agencies by enabling rapid screening of samples, prioritising laboratory testing and providing early warning of contamination risk. It can also be extended to real-time monitoring with IoT sensors.

9. Limitations and Scope

The dataset contains only nine parameters and does not include microbiological indicators (such as coliform bacteria) or heavy metals, which are critical for real-world safety assessment. The sample source, location and collection time are not specified, and the weak feature–target correlations may limit achievable accuracy. The model is intended as a screening aid and not a replacement for certified laboratory testing.

10. Conclusion

This project will apply machine learning to the Water Potability dataset to classify water as potable or non-potable and to estimate contamination risk. Through careful preprocessing, model comparison and interpretation of results, it aims to demonstrate how data-driven methods can complement traditional water testing and contribute to safer drinking water management.

11. References

- Kadiwal, A. Water Quality (Water Potability dataset), Kaggle.
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- Pedregosa, F. et al. Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2011.
- Chawla, N. V. et al. SMOTE: Synthetic Minority Over-sampling Technique. Journal of Artificial Intelligence Research, 16, 2002.